

KABALE UNIVERSITY

FACULTY OF COMPUTING

TEST ONE

NAME: ATUKWASE RODGERS
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COURSE: ADVANCED OBJECT ORIENTED PROGRAMMING
LECTURER: MR. BRIAN KASOZI

Classes

Animal (Parent class – Abstract): This demonstrates Abstraction

Dog (Child class – extends Animal): This demonstrates inheritance

Java program (Demonstrating OOP Concepts)

//Abstract class (Abstraction)

```
abstract class Animal {
```

```
String name;           //Attribute
```

```
// constructor
```

```
Animal (String name ) {
```

```
    this.name = name;
```

```
}
```

```
//Abstract method (must be implemented by child class)
```

```
abstract void sound();
```

//Normal method

Void eat() {

 System.out.println(name + “ is eating.”);

}

}

//Child class (Inheritance)

class Dog extends Animal {

int age; //Extra attribute

//Constructor

Dog (String name, int age){

 super (name); call parent constructor

 this.age = age;

}

//implement abstract method

@override

void sound() {

System.out.println (name + “ says: Woof!”);

}

// Extra method

void showAge(){

 System.out.println (name + “is “ + age + “years old.”);

}

}

```

//Main class
public class Main {
    public static void main (String[] args) {
        //Create object of Dog
        Dog d1 = new Dog("Buddy", 3);

        //Call methods
        d1.sound();           //From abstract method
        d1.eat();             //From parent class
        d1.showAge();         //From child class
    }
}

```

OUTPUT:

Buddy says: Woof!

Buddy is eating.

Buddy is 3 years old.

DESCRIPTION

Abstraction

Animal is abstract, hides details, and forces subclasses to implement sound().

Inheritance

Dog extends Animal and reuses its attributes/methods.

Attributes

name (in Animal), age(in Dog).

Methods

sound(), eat(), showAge().

Constructor

Both classes have constructors to initialize objects.

Objectives & Method Call

In Main, we create Dog d1 and call its methods.